

Experiment 02:

Study the basic network command and Network configuration commands like ping, variations of ip config, tracert, nslookup, netstat, arp, rarp, hostname, pathping and basic networking commands.

- ping
- ipconfig
- tracert
- nslookup
- netstat
- arp
- hostname
- pathping
- route
- net use

We will use a window 11 machine to understand these commands. Although some different OS like macOS & variants of Linux also follows the similar structure with different names but the working will be same.

1. What is Ping?

Ping is a network diagnostic command used to check whether another computer or network device is reachable over the network.

In other words,

Ping checks whether two devices can communicate with each other.

It also tells us:

- Is the destination device reachable?
- Is the network connection working?

- How much time does data take to travel?
- Were any packets lost during communication?

What Happens When We Run the Ping Command?

Many students think Ping simply sends a message saying "Hello."

That is **not correct**.

Ping works using a protocol called **ICMP (Internet Control Message Protocol)**.

What is ICMP?

ICMP (Internet Control Message Protocol) is a network protocol used to send control, error, and diagnostic messages between network devices.

Unlike HTTP or FTP,

ICMP is **not used for transferring files or web pages**.

It is mainly used for:

- Testing connectivity
- Reporting network errors
- Diagnosing network problems

The Ping command uses **ICMP Echo Request** and **ICMP Echo Reply** messages.

Eg: **8.8.8.8** is Google's Public DNS Server.

ping 8.8.8.8

```
Microsoft Windows [Version 10.0.26200.8655]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\goyal>ping google.com
```

```
Pinging google.com [2404:6800:4002:82a::200e] with 32 bytes of data:  
Reply from 2404:6800:4002:82a::200e: time=90ms  
Reply from 2404:6800:4002:82a::200e: time=62ms  
Reply from 2404:6800:4002:82a::200e: time=89ms  
Reply from 2404:6800:4002:82a::200e: time=121ms
```

```
Ping statistics for 2404:6800:4002:82a::200e:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 62ms, Maximum = 121ms, Average = 90ms
```

```
C:\Users\goyal>ping www.google.com
```

```
Pinging www.google.com [2001:4860:482b:7700::] with 32 bytes of data:  
Reply from 2001:4860:482b:7700::: time=89ms  
Reply from 2001:4860:482b:7700::: time=102ms  
Reply from 2001:4860:482b:7700::: time=107ms  
Reply from 2001:4860:482b:7700::: time=113ms
```

```
Ping statistics for 2001:4860:482b:7700:::  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 89ms, Maximum = 113ms, Average = 102ms
```

Ping Variations:

1. ping -t:

The ping -t command sends ICMP Echo Request packets continuously until the user manually stops the command.

In simple words,

ping -t keeps checking whether the destination device is reachable until you stop it yourself.

```
ping -t google.com
```

Press ctrl+C to stop.

```

C:\Users\goyal>ping -t 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=221ms TTL=109
Reply from 8.8.8.8: bytes=32 time=134ms TTL=109
Reply from 8.8.8.8: bytes=32 time=140ms TTL=109
Reply from 8.8.8.8: bytes=32 time=149ms TTL=109
Reply from 8.8.8.8: bytes=32 time=157ms TTL=109
Reply from 8.8.8.8: bytes=32 time=134ms TTL=109
Reply from 8.8.8.8: bytes=32 time=174ms TTL=109
Reply from 8.8.8.8: bytes=32 time=184ms TTL=109
Reply from 8.8.8.8: bytes=32 time=88ms TTL=109
Reply from 8.8.8.8: bytes=32 time=97ms TTL=109
Reply from 8.8.8.8: bytes=32 time=104ms TTL=109
Reply from 8.8.8.8: bytes=32 time=216ms TTL=109
Reply from 8.8.8.8: bytes=32 time=132ms TTL=109
Reply from 8.8.8.8: bytes=32 time=140ms TTL=109

Ping statistics for 8.8.8.8:
    Packets: Sent = 14, Received = 14, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 88ms, Maximum = 221ms, Average = 147ms
Control-C
^C

```

2. ping -n:

The ping -n command is used to specify the number of ICMP Echo Request packets that should be sent to the destination.

In simple words,

ping -n allows us to decide how many Ping requests Windows should send.

```
ping -n 20 192.168.1.1
```

```

C:\Users\goyal>ping -n 2 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=217ms TTL=109
Reply from 8.8.8.8: bytes=32 time=122ms TTL=109

Ping statistics for 8.8.8.8:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 122ms, Maximum = 217ms, Average = 169ms

```

3. ping -l:

The -l option is used to specify the size of the data payload (in bytes) that is sent in each ICMP Echo Request packet.

```
ping -l 100 google.com
```

```
C:\Users\goyal>ping -l 54 www.google.com

Pinging www.google.com [2001:4860:482c:7700::] with 54 bytes of data:
Reply from 2001:4860:482c:7700::: time=109ms
Reply from 2001:4860:482c:7700::: time=101ms
Reply from 2001:4860:482c:7700::: time=209ms
Reply from 2001:4860:482c:7700::: time=75ms

Ping statistics for 2001:4860:482c:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 75ms, Maximum = 209ms, Average = 123ms
```

4. ping -f:

Fragmentation is the process of dividing a large IP packet into smaller packets so that it can travel through a network with a smaller Maximum Transmission Unit (MTU).

MTU tells us the maximum packet size a network can carry in one piece.

The letter **f** stands for:

Don't Fragment

It tells Windows:

"Send the packet, but do NOT allow it to be fragmented."

The -f option sets the "Don't Fragment (DF)" flag in the IP header. If the packet is too large for any link along the path, it is discarded instead of being fragmented.

```
ping -f -l 1472 google.com
```

```
C:\Users\goyal>ping -f -l 1472 8.8.8.8

Pinging 8.8.8.8 with 1472 bytes of data:
Reply from 10.223.123.153: Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
```

5. ping -a:

The -a option resolves an IP address into its hostname (computer name), if the name can be resolved.

In simple English,

ping -a tries to find the name of the computer from its IP address.

```
ping -a 192.168.1.10
```

2. What is IPCONFIG?

IPCONFIG (Internet Protocol Configuration) is a Windows command-line utility used to view and manage the network configuration of a computer.

1. ipconfig

```
C:\Users\goyal>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40d1:d:91b2:1ac5:dab1:994f:63dd
    Temporary IPv6 Address. . . . . : 2409:40d1:d:91b2:1b1:a1df:17bd:a88d
    Link-local IPv6 Address . . . . . : fe80::5b47:fece:3721:502e%17
    IPv4 Address. . . . . : 10.223.123.16
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6432:adff:fe4f:51de%17
                                10.223.123.153

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a935:f712:2081:2ebf%53
    IPv4 Address. . . . . : 172.19.144.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :
```

2. Ipconfig /all

ipconfig /all displays the complete TCP/IP configuration of all network adapters installed on the computer.

```
C:\Users\goyal>ipconfig -all

Windows IP Configuration

Host Name . . . . . : Devengoyal
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : C2-35-32-87-F1-71
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : C6-35-32-87-F1-71
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Fortinet Virtual Ethernet Adapter (NDIS 6.30)
Physical Address. . . . . : 00-09-0F-FE-00-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. . . . . : C8-35-32-87-F1-71
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2409:40d1:d:91b2:1ac5:dab1:994f:63dd(Preferred)
Temporary IPv6 Address. . . . . : 2409:40d1:d:91b2:1b1:aldf:17bd:a88d(Preferred)
Link-local IPv6 Address . . . . . : fe80::5b47:fece:3721:502e%17(Preferred)
IPv4 Address. . . . . : 10.223.123.16(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 15 July 2026 9.30.28 AM
Lease Expires . . . . . : 15 July 2026 11.30.31 AM
Default Gateway . . . . . : fe80::6432:adff:fe4f:51de%17
                          10.223.123.153
DHCP Server . . . . . : 10.223.123.153
DHCPv6 IAID . . . . . : 180368690
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-04-73-6E-C0-35-32-87-F1-71
DNS Servers . . . . . : 10.223.123.153
                          2409:40d1:d:91b2::2b
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-7E-0C-DF
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a935:f712:2081:2ebf%53(Preferred)
IPv4 Address. . . . . : 172.19.144.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 889197917
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-04-73-6E-C0-35-32-87-F1-71
NetBIOS over Tcpip. . . . . : Enabled
```

3. ipconfig /release

ipconfig /release releases (returns) the current IPv4 address obtained from the DHCP server.

```

C:\Users\goyal>ipconfig /release

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Ethernet 2 while it has its media disconnected.

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40d1:d:91b2:1ac5:dab1:994f:63dd
    Temporary IPv6 Address. . . . . : 2409:40d1:d:91b2:1b1:a1df:17bd:a88d
    Link-local IPv6 Address . . . . . : fe80::5b47:fece:3721:502e%17
    Default Gateway . . . . . : fe80::6432:adff:fe4f:51de%17

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a935:f712:2081:2ebf%53
    IPv4 Address. . . . . : 172.19.144.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :

```

4. ipconfig /renew:

It requests a **new IP address from the DHCP server**, and together these two commands form one of the most common troubleshooting procedures used by network engineers.

```

C:\Users\goyal>ipconfig /renew

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
No operation can be performed on Ethernet 2 while it has its media disconnected.

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40d1:d:91b2:1ac5:dab1:994f:63dd
    Temporary IPv6 Address. . . . . : 2409:40d1:d:91b2:1b1:a1df:17bd:a88d
    Link-local IPv6 Address . . . . . : fe80::5b47:fece:3721:502e%17
    IPv4 Address. . . . . : 10.223.123.16
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6432:adff:fe4f:51de%17
                                10.223.123.153

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a935:f712:2081:2ebf%53
    IPv4 Address. . . . . : 172.19.144.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :

```


3. What is Tracert?

Imagine you are opening www.google.com, but the website is loading very slowly.

You may ask:

- Is my Wi-Fi working properly?
- Is my ISP (Internet Service Provider) having a problem?
- At which router is the delay occurring?
- Is the destination server reachable?
- Through which routers is my data travelling?

A simple ping command only tells us:

- Whether the destination is reachable.
- The response time.

But it **does not tell us the path** taken by the data.

This is where the **tracert** command becomes useful.

It shows **every router (hop)** through which the packet travels before reaching the destination.

tracert stands for **Trace Route**

It is a Windows command used to trace the complete path taken by data packets from the source computer to the destination computer.

It displays:

- Every intermediate router (Hop)
- Response time of each hop
- IP address or hostname of each router

- Whether any router is delaying or dropping packets

Example:

```
</> cmd

tracert 8.8.8.8
```

Output:

```
C:\>tracert google.com

Tracing route to google.com [142.250.183.238]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms    192.168.1.1
  2     10 ms    11 ms    10 ms    10.20.0.1
  3     18 ms    17 ms    18 ms    172.16.20.1
  4     25 ms    24 ms    25 ms    103.21.45.10
  5     31 ms    30 ms    31 ms    142.250.183.238

Trace complete.
```

```
C:\Users\goyal>tracert www.google.com

Tracing route to www.google.com [2001:4860:4826:7700::]
over a maximum of 30 hops:

  1     12 ms     5 ms     4 ms    2409:40d1:d:91b2::2b
  2      *        *        *      Request timed out.
  3    187 ms    101 ms    99 ms    2405:200:520b:21:3925::ff06
  4    110 ms     56 ms    144 ms    2405:200:872:1513:61::6
  5      *        *        *      Request timed out.
  6    126 ms    100 ms    100 ms    2405:200:801:4000::336
  7      *        *        *      Request timed out.
  8      *        *        *      Request timed out.
  9     69 ms    143 ms    100 ms    2001:4860:1:1::f48
 10    110 ms    101 ms    100 ms    2001:4860:4826:7700::

Trace complete.
```

On **Windows**:

- tracert uses **ICMP Echo Request** packets with increasing TTL values.

On many **Linux/Unix** systems:

- traceroute traditionally uses **UDP** packets by default (though it can also use ICMP or TCP depending on options).

Variations of tracert:

1. `tracert -d destination`: Does not resolve host names. Shows only IP addresses, making the command faster.

```
C:\Users\goyal>tracert -d
A target name or address must be specified.

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
  -d          Do not resolve addresses to hostnames.
  -h maximum_hops  Maximum number of hops to search for target.
  -j host-list  Loose source route along host-list (IPv4-only).
  -w timeout    Wait timeout milliseconds for each reply.
  -R          Trace round-trip path (IPv6-only).
  -S srcaddr    Source address to use (IPv6-only).
  -4          Force using IPv4.
  -6          Force using IPv6.

C:\Users\goyal>tracert -d www.google.com

Tracing route to www.google.com [2001:4860:4826:7700::]
over a maximum of 30 hops:

  1    10 ms    4 ms    6 ms  2409:40d1:d:91b2::2b
  2    *        *        *      Request timed out.
  3   272 ms   217 ms   209 ms  2405:200:520b:21:3925::ff06
  4   169 ms   101 ms   100 ms  2405:200:872:1513:61::6
  5    *        *        *      Request timed out.
  6    86 ms   102 ms    98 ms  2405:200:801:4000::336
  7    *        *        *      Request timed out.
  8    *        *        *      Request timed out.
  9   182 ms   203 ms   101 ms  2001:4860:1:1::f48
 10   103 ms   101 ms    80 ms  2001:4860:4826:7700::

Trace complete.
```

2. `tracert -h <max_hops> destination`: Specifies the maximum number of hops to search.

```
C:\Users\goyal>tracert -h 5 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 5 hops:

  1    14 ms    5 ms    7 ms  10.223.123.153
  2     8 ms    7 ms    9 ms  192.0.0.1
  3    *        *        *      Request timed out.
  4    *        *        *      Request timed out.
  5    *        *        *      Request timed out.

Trace complete.

C:\Users\goyal>nslookup google.com
Server:      Unknown
Address:  10.223.123.153

Non-authoritative answer:
Name:      google.com
Addresses: 2404:6800:4013:813::66
           2404:6800:4013:813::64
           2404:6800:4013:813::8b
           2404:6800:4013:813::65
           142.250.29.113
           142.250.29.139
           142.250.29.101
           142.250.29.138
```

3. `tracert -?`: Displays help for the tracert command.

4. What is nslookup?

Many of us know websites by names like:

- www.google.com
- www.youtube.com
- www.amazon.in

But computers **do not communicate using names**. They communicate using **IP addresses**.

The job of **DNS** is to convert a **domain name into an IP address** (and sometimes an IP address back into a domain name).

The nslookup command allows us to check whether the DNS server is working correctly and to find the IP address of a domain name.

What is the requirement of nslookup?

Imagine you type:

www.google.com

into your web browser.

Your browser cannot directly communicate with the name **google.com**.

It first asks a **DNS Server**:

"What is the IP address of google.com?"

The DNS server replies with something like:

142.250.183.238

Only after getting the IP address does your computer connect to Google's server.

If the DNS server is not working:

- Websites may not open.
- Internet may appear to be down, even if the network connection is fine.
- You may be able to ping an IP address, but not a website name.

This is why the nslookup command is very useful.

Example:

```
</> cmd

nslookup google.com
```

Output:

```
C:\>nslookup google.com

Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Addresses: 142.250.183.238
          2404:6800:4007:80b::200e
```

```
C:\Users\goyal>nslookup google.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4013:811::8a
          2404:6800:4013:811::71
          2404:6800:4013:811::64
          2404:6800:4013:811::66
          192.178.134.101
          192.178.134.113
          192.178.134.139
          192.178.134.102
          192.178.134.138
          192.178.134.100
```

Non-Authoritative Answer means, The DNS server is giving the answer from its **cache**, not directly from the original authoritative DNS server for that domain.

Addresses: 142.250.183.238 (IPv4)
2404:6800:4007:80b::200e (IPv6)

Command	Purpose	Example
nslookup google.com	Find the IP address of a web:	nslookup google.com
nslookup 8.8.8.8	Reverse lookup (IP to doma	nslookup 8.8.8.8

nslookup google.com 8.8.8.8	Use a specific DNS server	nslookup google.com 8.8.8.8
nslookup -type=A google.com	Show IPv4 records	nslookup -type=A google.com
nslookup -type=AAAA google.com	Show IPv6 records	nslookup -type=AAAA google.com

```

C:\Users\goyal>nslookup -type=A 8.8.8.8
Server:  UnKnown
Address:  10.223.123.153

Name:     dns.google
Address:  8.8.8.8

C:\Users\goyal>nslookup -type=AAAA 8.8.8.8
Server:  UnKnown
Address:  10.223.123.153

Name:     dns.google
Address:  8.8.8.8

```

5. What is netstat?

netstat stands for **Network Statistics**

It is a command-line utility used to display information about:

- Active network connections
- Listening ports
- Routing information
- Network statistics
- Protocol statistics

Network administrators use this command to monitor network activity and troubleshoot network-related problems

Why Do We Need netstat?

Imagine your Internet is running very slowly.

Possible reasons:

- Too many applications are using the Internet.
- A malicious program is secretly connected to another computer.
- A service is listening on an unwanted port.
- An application has too many open connections.

Instead of guessing,

netstat allows us to see all current network activity.

Example:

```
</> cmd  
  
netstat
```

Output:

```
</> cmd  
  
C:\>netstat  
  
Active Connections  
  
Proto Local Address Foreign Address State  
TCP 192.168.1.10:49725 142.250.183.238:443 ESTABLISHED  
TCP 192.168.1.10:49730 13.107.42.16:443 ESTABLISHED  
TCP 127.0.0.1:49675 127.0.0.1:49676 ESTABLISHED
```

C:\Users\goyal>netstat

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.223.123.16:58946	38.134.110.236:https	SYN_SENT
TCP	10.223.123.16:60573	69.173.158.64:https	ESTABLISHED
TCP	10.223.123.16:61943	38.134.110.236:https	SYN_SENT
TCP	127.0.0.1:65361	kubernetes:65361	ESTABLISHED
TCP	127.0.0.1:65362	kubernetes:65361	ESTABLISHED
TCP	127.0.0.1:65363	kubernetes:65364	ESTABLISHED
TCP	127.0.0.1:65364	kubernetes:65363	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:49309	[2a04:4e42:fe7::810]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:49766	[2405:200:1609:1731::312c:8abb]:https	CLOSE_WAIT
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50144	lb-140-82-114-25-1ad:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50206	[64:ff9b::2686:6eeb]:https	SYN_SENT
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50279	[2603:1061:14:153::1]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50377	server-13-225-103-68:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50385	[2606:4700:90c2:7cbc:c264:560:5ff2:75c4]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50895	[2620:1ec:33:1::12]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:51052	[2606:4700:90c1:9e54:7564:55e:6150:384c]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:51419	208:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:51557	[64:ff9b::45ad:9e40]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:52769	[2620:1ec:33::10]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:52819	[2405:200:1630:1200::312c:b389]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53007	[64:ff9b::481c:7b6b]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53068	[2606:4700:90c3:ce32:c50c:560:2d49:6da5]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53174	[2600:1901:0:d5ad:]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53401	[2603:1040:a06:6::]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53676	63:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:55422	[2405:200:1630:1200::312c:b398]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:56380	server-108-158-245-108:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:56560	[2620:1ec:33:11]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:57112	whatsapp-cdn6-shv-01-del3:https	TIME_WAIT
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:57328	[64:ff9b::9765:6a85]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:57599	[64:ff9b::45ad:9e40]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:57919	[64:ff9b::2686:6eeb]:https	SYN_SENT
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:58337	[2606:4700:3037::ac43:ce59]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:58637	server-108-158-245-83:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59036	[2405:200:1630:1200::312c:b388]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59415	[2a04:4e42:19::644]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59429	[64:ff9b::acbc:9b19]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59448	[2603:1020:206:10::e0]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59579	[2a04:4e42:fea::810]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:60110	[2405:200:1630:1200::312c:b398]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:60345	ec2-18-97-36-75:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61040	[2606:4700:90c3:ce32:c5a8:55e:2d49:6da5]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61442	server-18-164-246-127:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61660	ec2-54-254-212-173:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61741	ec2-47-130-218-206:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61766	[64:ff9b::284f:8d98]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:62787	34:https	TIME_WAIT
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:62826	[2620:1ec:33:10]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:63768	[64:ff9b::43c7:9657]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:64392	server-108-158-232-20:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:64731	[2600:9000:257b:2e00:f:cc3f:6c80:93a1]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:64876	[2620:1ec:33:11]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:65344	[2406:da18:444:1e1e:e8d1:db4c:792e:20e6]:https	ESTABLISHED
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:65445	[2606:4700:9641:d8b7:9a64:561:ccf5:ac50]:https	ESTABLISHED

Command	Purpose	Example
netstat	Show active TCP connections	netstat
netstat -a	Show all connections and listening ports	netstat -a
netstat -n	Show numeric IP addresses and port numbers	netstat -n
netstat -o	Show Process IDs (PID)	netstat -o
netstat -e	Show Ethernet statistics	netstat -e

netstat -r	Show routing table	netstat -r
netstat -p TCP	Show only TCP connections	netstat -p TCP
netstat -p UDP	Show only UDP connections	netstat -p UDP
netstat -s	Show protocol statistics	netstat -s
netstat -ano	Show all connections with numeric addresses and PID	netstat -ano
netstat -?	Show help	netstat -?

```
C:\Users\goyal>netstat -o
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	10.223.123.16:60573	69.173.158.64:https	ESTABLISHED	30104
TCP	127.0.0.1:65361	kubernetes:65362	ESTABLISHED	5524
TCP	127.0.0.1:65362	kubernetes:65361	ESTABLISHED	5524
TCP	127.0.0.1:65363	kubernetes:65364	ESTABLISHED	5524
TCP	127.0.0.1:65364	kubernetes:65363	ESTABLISHED	5524
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:49309	[2a04:4e42:fe7::810]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50144	lb-140-82-114-25-1ad:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50377	server-13-225-103-68:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50385	[2606:4700:90c2:7cbc:c264:560:5ff2:75c4]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:50895	[2620:1ec:33:1::12]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:51052	[2606:4700:90c1:9e54:7564:55e:6150:384c]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:51419	208:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:51557	[64:ff9b::45ad:9e40]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:52769	[2620:1ec:33::10]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:52819	[2405:200:1630:1200::312c:b389]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53047	[64:ff9b::481c:7b6b]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53068	[2606:4700:90c3:ce32:c50c:560:2d49:6da5]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53174	[2600:1901:0:d5ad::]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53401	[2603:1040:a06:6::]:https	ESTABLISHED	5216
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:53676	63:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:55422	[2405:200:1630:1200::312c:b398]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:56161	[2606:4700:90c1:d8b7:9aa8:562:ccf5:ac50]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:56380	server-108-158-245-108:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:56560	[2620:1ec:33::11]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:57328	[64:ff9b::9765:6a85]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:57599	[64:ff9b::45ad:9e40]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:58337	[2606:4700:3037::ac43:ce59]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:58637	server-108-158-245-83:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59036	[2405:200:1630:1200::312c:b388]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59415	[2a04:4e42:19::644]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59429	[64:ff9b::acbc:9b19]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59448	[2603:1020:206:10::e0]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:59579	[2a04:4e42:fea::810]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:60110	[2405:200:1630:1200::312c:b398]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:60345	ec2-18-97-36-75:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61040	[2606:4700:90c3:ce32:c5a8:55e:2d49:6da5]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61442	server-18-164-246-127:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61660	ec2-54-254-212-173:https	CLOSE_WAIT	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61741	ec2-47-130-218-206:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:61766	[64:ff9b::284f:8d98]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:62787	34:https	TIME_WAIT	0
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:62826	[2620:1ec:33::10]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:63768	[64:ff9b::43c7:9657]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:64392	server-108-158-232-20:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:64731	[2600:9000:257b:2e00:f:cc3f:6c80:93a1]:https	ESTABLISHED	30104
TCP	[2409:40d1:d:91b2:1b1:aldf:17bd:a88d]:64876	[2620:1ec:33::11]:https	ESTABLISHED	30104

6. What is arp?

This command is one of the most important commands because it directly relates to the **Data Link Layer (Layer 2)** and the **Network Layer (Layer 3)**.

It helps us to understand one of the most confusing concepts in networking:

"How does a computer know the MAC Address of another computer if it only knows its IP Address?"

The answer is:

ARP (Address Resolution Protocol): The arp command is used to view and manage the ARP Cache, which contains mappings between IP addresses and MAC addresses.

Suppose your computer wants to send data to another computer on the same LAN.

Your computer knows:

Destination IP Address: 192.168.0.50

But Ethernet communication cannot happen using only an IP address.

Ethernet frames require:

Destination MAC Address

Now the question is:

How will the sender know the MAC Address corresponding to 192.168.0.50?

The answer is:

Using **ARP**.

The arp command allows us to view and manage this IP-to-MAC mapping stored on our computer.

Examples:

```
</> cmd
```

```
arp -a
```

Display all the entries stored inside the ARP Cache.

Output:

```
</> cmd

Interface: 192.168.0.197 --- 0x8

Internet Address      Physical Address      Type
192.168.0.1           20-47-47-AA-BB-CC     dynamic
192.168.0.15          34-97-F6-22-12-45     dynamic
192.168.0.50          60-45-CB-55-11-22     dynamic
```

What does this 0x8 means?

Interface Index

Windows assigns every network adapter a unique identification number.

Example

Interface	Interface Index
Wi-Fi	0x8
Ethernet	0x9
VPN	0x12

```
</> cmd

arp -d 192.168.0.15
```

Delete an entry from the ARP Cache.

```
</> cmd

arp -d *
```

Delete all entries

```
</> cmd  
  
arp -s 192.168.0.15 00-15-5D-45-12-AB
```

Create a **Static ARP Entry**.

Means, windows permanently stores:
192.168.0.15



00-15-5D-45-12-AB

instead of learning it automatically.

Administrator privilege is required.

```
C:\Users\goyal>arp -a  
  
Interface: 10.223.123.16 --- 0x11  
Internet Address      Physical Address      Type  
10.223.123.153        66-32-ad-4f-51-de     dynamic  
10.223.123.255        ff-ff-ff-ff-ff-ff     static  
224.0.0.22            01-00-5e-00-00-16     static  
224.0.0.251          01-00-5e-00-00-fb     static  
224.0.0.252          01-00-5e-00-00-fc     static  
239.255.255.250       01-00-5e-7f-ff-fa     static  
255.255.255.255       ff-ff-ff-ff-ff-ff     static  
  
Interface: 172.19.144.1 --- 0x35  
Internet Address      Physical Address      Type  
172.19.146.170        00-15-5d-31-50-9e     dynamic  
172.19.159.255        ff-ff-ff-ff-ff-ff     static  
224.0.0.22            01-00-5e-00-00-16     static  
224.0.0.251          01-00-5e-00-00-fb     static  
224.0.0.252          01-00-5e-00-00-fc     static  
239.255.255.250       01-00-5e-7f-ff-fa     static  
  
C:\Users\goyal>arp -d 8.8.8.8  
The ARP entry deletion failed: The requested operation requires elevation.
```

7. What is hostname?

The hostname command is used to display the name of the current computer.

How to change hostname: press win + r -> type sysdm.cpl

Over the network we follow a order:

Useful Tip

Order of commands;

hostname

Shows **Who am I?**

ipconfig

Shows **What is my IP Address?**

arp -a

Shows **What MAC Addresses have I learned on my local network?**

```
C:\Users\goyal>hostname  
Devengoyal
```

8. What is pathping?

Pathping = Ping + Tracert

It combines the features of both commands.

Suppose a user complains:

"Google opens very slowly."

Now, you need to answer:

- Is the destination reachable?
- Which route is the packet taking?
- Which router is causing the delay?
- At which router are packets being lost?

Neither ping nor tracert alone can answer all these questions.

The pathping command provides all this information in a single command.

Path Ping

It is a Windows networking command that combines the functionality of:

- ping
- tracert

It traces the route to the destination and then sends multiple ping packets to **each router (hop)** to calculate:

- Packet Loss
- Response Time
- Network Delay

In simple terms, **The pathping command traces the path to a destination and analyzes packet loss and latency at each router along the route.**

Why Do We Need pathping?

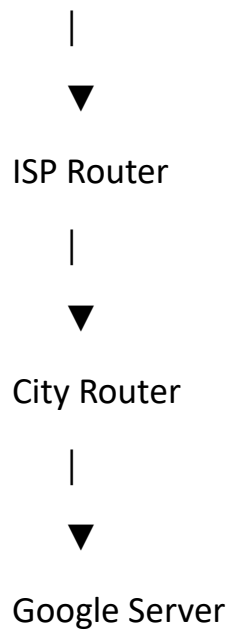
Imagine a route like this:

Your Laptop

|



Home Router



Suppose the **ISP Router** is overloaded.

Result:

- Some packets are dropped.
- Internet becomes slow.

A normal ping only tells you:

Reply from Google

A tracert only tells you:

The packet passed through ISP Router.

But **pathping** tells you:

Packet Loss at ISP Router = 18%

Now you know exactly where the problem is.

```
</> cmd  
  
pathping google.com
```

Output:


```
</> cmd
```

```
C:\>pathping google.com
```

```
Tracing route to google.com [142.250.183.238]
```

```
0  YOUR-PC
1  192.168.0.1
2  10.20.0.1
3  172.16.20.1
4  142.250.183.238
```

```
Computing statistics for 100 seconds...
```

Source to Here Hop	Lost/Sent	This Node/Link Lost/Sent	Address
0	0/100	0/100	YOUR-PC
1	0/100	0/100	192.168.0.1
2	2/100	2/100	10.20.0.1
3	5/100	3/100	172.16.20.1
4	5/100	0/100	google.com

```
Trace complete.
```

Each hop represents one router between the source and the destination.

Example:

Hop	Device
0	Your Computer
1	Home Router
2	ISP Router
3	Internet Router
4	Google Server

```

C:\Users\goyal>pathping 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:
 0 Devengoyal [10.223.123.16]
 1 10.223.123.153
 2 192.0.0.1
 3 * * *
Computing statistics for 50 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
 0                               0/ 100 = 0%      Devengoyal [10.223.123.16]
 1  13ms      0/ 100 = 0%      0/ 100 = 0%      10.223.123.153
 2  ---      100/ 100 =100%   100/ 100 =100%   |
 3  ---      100/ 100 =100%   0/ 100 = 0%      192.0.0.1

Trace complete.

```

Lost/Sent (Source to Here)

Example

2/100

Meaning:

Out of **100 packets sent**, **2 packets were lost** before reaching this hop.

Packet Loss:

2%

Lost/Sent (This Node/Link)

Example

3/100

This tells how many packets were lost **at that specific router or link**.

This helps identify the exact location of the problem.

Address

Shows the IP address or Host Name of the router.

Example

192.168.0.1

Your Home Router.

10.20.0.1

ISP Router.

google.com

Destination Server.

What is Packet Loss?

Packet Loss means that some packets sent by your computer never reached their destination.

Example:

You send:

100 Packets

Destination receives:

95 Packets

Then,

Packet Loss = 5%

Formula

Packet Loss (%) = (Packets Lost / Packets Sent) × 100

Example

Packets Sent:

100

Packets Lost:

5

Calculation:

Packet Loss = $(5 / 100) \times 100$

Packet Loss = 5%

Why Does Packet Loss Occur?

Common reasons:

- Network congestion.
- Faulty cables.
- Weak Wi-Fi signal.
- Router overload.
- Hardware failure.
- ISP issues.

Which Protocol Does pathping Use?

pathping uses **ICMP (Internet Control Message Protocol)**.

It sends:

- ICMP Echo Requests.
- ICMP Echo Replies.

Just like ping, but it also traces the route like tracert.

Practical Uses

Network administrators use pathping to:

- Detect packet loss.
- Identify slow routers.
- Troubleshoot network performance.
- Verify routing paths.
- Diagnose ISP-related issues.